DBMS Interview Question Bank

1. Core DBMS & Database Fundamentals

Q1	What is DBMS?		
Ans	A DBMS (Database Management System) is software that helps store, manage, and manipulate data efficiently in a structured way.		
Co.	TCS • Infosys • Wipro • Accenture • Capgemini	BL	L1 – Remember
Ex.	□ MySQL, Oracle, PostgreSQL, MS SQL Server are popular DBMS products.		

Q2	What is RDBMS?		
Ans	An RDBMS (Relational Database Management System) organizes data in tables (relations) using rows and columns, enforcing relationships via keys.		
Co.	TCS • Infosys • HCL • Amazon	BL	L1 – Remember
Ex.	□ MySQL, Oracle, SQL Server — all store data in rows/columns and use foreign keys to link tables.		

Q3	Difference between DBMS and RDBMS?		
Ans	DBMS stores data as files; RDBMS uses relational tables and enforces keys, constraints, and relationships between tables.		
Co.	Wipro • Accenture • Cognizant • Capgemini	BL	L2 – Understand
Ex.	□ A simple file-based inventory system = DBMS. A payroll system with Employees↔Departments tables = RDBMS.		

Q4	What is a database?		
Ans	A database is a structured collection of data stored and accessed electronically, organized to allow efficient retrieval and manipulation.		
Co.	TCS • Infosys • Accenture	BL	L1 – Remember
Ex.	□ A university database holds students, courses, grades — all structured and related.		

Q5	What is a schema?		
Ans	A logical structure defining how data is organized — tables, fields, data types, constraints, and relationships.		
Co.	Amazon • Google • Microsoft • Flipkart	BL	L1 – Remember
Ex.	□ A schema for an e-commerce DB defines tables like Orders(order_id, user_id, total), Products(product_id, name, price).		

Q6	Difference between schema and database?

Ans	A database is the physical collection of data stored on disk; a schema is the logical structure/blueprint that defines how that data is organized.		
Co.	**Oracle • IBM • SAP • TCS**	BL	**L2 – Understand**
Ex.	`A hospital database may have multiple schemas: patient_schema, billing_schema, pharmacy_schema.`		

Q7	What is a relation?		
Ans	A table with rows (tuples) and columns (attributes) in a relational database. Each relation represents a real-world entity.		
Co.	**TCS • Infosys**	BL	**L1 – Remember**
Ex.	`Employee(emp_id, name, salary) is a relation.`		

Q8	What is an entity?		
Ans	An object in the real world that is representable in a database as a table — something about which data needs to be stored.		
Co.	**Accenture • Cognizant • Wipro**	BL	**L1 – Remember**
Ex.	`Student, Course, Department, Employee are entities.`		

Q9	What is a weak entity?		
Ans	An entity that cannot be uniquely identified by its own attributes alone and depends on a strong (parent) entity for identification.		
Co.	**Amazon • Oracle • IBM**	BL	**L2 – Understand**
Ex.	`OrderItem is a weak entity — it needs OrderID from the Order (strong entity) to be unique.`		

Q10	What is data integrity?		
Ans	The accuracy, consistency, and reliability of data throughout its lifecycle — ensured by constraints, transactions, and validation rules.		
Co.	**TCS • Infosys • Wipro • HCL**	BL	**L2 – Understand**
Ex.	`A NOT NULL constraint on email ensures every user has an email, maintaining integrity.`		

2. Keys & Constraints

Q11	What is a primary key?		
Ans	A unique identifier for each record in a table; cannot contain NULL values and must be unique across all rows.		
Co.	**TCS • Infosys • Wipro • Cognizant • Amazon**	BL	**L1 – Remember**
Ex.	`In Employees(emp_id, name, salary), emp_id is the primary key — no two employees share the same emp_id.`		

Q12	What is a foreign key?		
Ans	A field in one table that references the primary key of another table, establishing a link between the two tables.		
Co.	TCS • Infosys • HCL • Oracle	BL	L1 – Remember
Ex.	□ In `Orders(order_id, customer_id)`, `customer_id` is a FK referencing `Customers(customer_id)`.		

Q13	What is a candidate key?		
Ans	A field or combination of fields that can uniquely identify a record; one candidate key is chosen as the primary key, the rest become alternate keys.		
Co.	Amazon • Google • Flipkart • TCS	BL	L2 – Understand
Ex.	□ In `Employee(emp_id, email, name)`, both emp_id and email can uniquely identify a row — both are candidate keys.		

Q14	What is a composite key?		
Ans	A primary key consisting of two or more columns combined to uniquely identify a record when no single column is sufficient alone.		
Co.	TCS • Wipro • Cognizant	BL	L2 – Understand
Ex.	□ In `Enrollment(student_id, course_id, semester)`, the combination of all three forms the composite key.		

Q15	What is a surrogate key?		
Ans	An artificial, system-generated key (usually auto-increment integer) used to uniquely identify a record, not derived from real-world data.		
Co.	Amazon • Microsoft • Oracle • SAP	BL	L2 – Understand
Ex.	□ AUTO_INCREMENT emp_id in MySQL is a surrogate key — it has no business meaning, just uniqueness.		

Q16	Difference between natural key and surrogate key?		
Ans	A natural key comes from existing data attributes (e.g., SSN, email); a surrogate key is system-generated and has no real-world meaning.		
Co.	Oracle • IBM • SAP	BL	L2 – Understand
Ex.	□ Email as PK = natural key. Auto-increment ID = surrogate key. Surrogate keys are safer as emails can change.		

Q17	What is a Super Key? Difference from Candidate Key?		
Ans	A super key is any set of attributes that uniquely identifies a tuple. A candidate key is the minimal super key — no subset of a candidate key can also uniquely identify a record.		
Co.	Amazon • Google • Microsoft • Adobe	BL	L4 – Analyse
Ex.	□ {emp_id, name} is a super key but not a candidate key (we can remove 'name'). {emp_id} alone is a candidate key.		

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Q18	What is an Alternate Key?		
Ans	All candidate keys except the one chosen as the Primary Key are called alternate keys. They can still uniquely identify records.		
Co.	TCS • Infosys • Wipro	BL	L2 – Understand
Ex.	`□ If both emp_id and email are candidate keys and emp_id is chosen as PK, then email becomes the alternate key.`		

Q19	What is referential integrity?		
Ans	Ensures that a foreign key value always points to a valid existing primary key in the referenced table — no orphan records allowed.		
Co.	TCS • Oracle • IBM • SAP	BL	L2 – Understand
Ex.	`□ If Orders.customer_id = 5, then customer_id=5 must exist in Customers table, or the FK constraint is violated.`		

Q20	What is a constraint?		
Ans	Rules applied to table columns to maintain data integrity. Common types: NOT NULL, UNIQUE, CHECK, PRIMARY KEY, FOREIGN KEY, DEFAULT.		
Co.	TCS • Infosys • Cognizant	BL	L1 – Remember
Ex.	`□ CHECK(salary > 0) ensures no negative salary is inserted. NOT NULL on email makes it mandatory.`		

Q21	What is a cascade delete/update?		
Ans	Automatically deletes or updates child (referencing) records when the parent (referenced) record is deleted or updated.		
Co.	Oracle • Microsoft • SAP • Amazon	BL	L3 – Apply
Ex.	`□ ON DELETE CASCADE on Orders means deleting a Customer also deletes all their Orders automatically.`		

Q22	What is a foreign key constraint?		
Ans	Ensures that a column value in the child table matches a primary key value in the parent table, maintaining referential integrity.		
Co.	TCS • Wipro • Oracle • IBM	BL	L2 – Understand
Ex.	`□ ALTER TABLE Orders ADD CONSTRAINT fk_customer FOREIGN KEY (customer_id) REFERENCES Customers(customer_id);`		

Q23	Difference between primary key and unique key?		
Ans	Primary key: uniquely identifies rows, cannot be NULL, only one per table. Unique key: ensures uniqueness, can have one NULL per column, multiple per table.		
Co.	TCS • Infosys • Amazon • Flipkart	BL	L2 – Understand
Ex.	`□ emp_id = PK (no NULLs). email has UNIQUE constraint (can be NULL once if someone hasn't provided it).`		

3. SQL Commands & Query Basics

Q24	What is SQL?		
Ans	SQL (Structured Query Language) is a standard language to interact with relational databases — to create, read, update, and delete data.		
Co.	TCS • Infosys • Wipro • Accenture • HCL • Amazon • Flipkart	BL	L1 – Remember
Ex.	`SELECT * FROM Employees WHERE salary > 50000;`		

Q25	What are the types of SQL commands?		
Ans	DDL (Data Definition: CREATE, ALTER, DROP), DML (Data Manipulation: INSERT, UPDATE, DELETE), DCL (Data Control: GRANT, REVOKE), TCL (Transaction Control: COMMIT, ROLLBACK, SAVEPOINT).		
Co.	TCS • Infosys • Wipro • Oracle • Cognizant	BL	L1 – Remember
Ex.	`DDL: CREATE TABLE. DML: INSERT INTO. DCL: GRANT SELECT ON table TO user. TCL: COMMIT;`		

Q26	What is NULL in SQL?		
Ans	NULL represents missing, unknown, or undefined data. It is not equivalent to zero (0) or an empty string ('').		
Co.	TCS • Infosys • Amazon • Microsoft	BL	L1 – Remember
Ex.	`SELECT * FROM Employees WHERE manager_id IS NULL; — finds employees with no manager.`		

Q27	Difference between IS NULL and = NULL?		
Ans	Use IS NULL to check for NULL values; '= NULL' always evaluates to UNKNOWN in SQL and never returns rows.		
Co.	TCS • Infosys • Wipro • Oracle	BL	L2 – Understand
Ex.	`SELECT * FROM Employees WHERE manager_id IS NULL; ✓ | SELECT * WHERE manager_id = NULL; ✗ (returns nothing)`		

Q28	What is COALESCE()?		
Ans	Returns the first non-null value in a list of expressions. Useful for handling NULLs in queries.		
Co.	Amazon • Microsoft • Oracle • Flipkart	BL	L3 – Apply
Ex.	`SELECT COALESCE(phone, email, 'No Contact') AS contact FROM Employees; — returns first non-null value.`		

Q29	Difference between DELETE and TRUNCATE?		
Ans	DELETE removes specific rows (with WHERE clause), can be rolled back, and fires triggers. TRUNCATE removes all rows, resets identity/auto-increment, and cannot be rolled back (in most DBs).		
Co.	TCS • Infosys • Amazon • Oracle • Wipro	BL	L2 – Understand
Ex.	`DELETE FROM Employees WHERE emp_id=5; — removes one row. TRUNCATE TABLE Employees; — removes all rows instantly.`		

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Q30	Difference between DELETE, TRUNCATE, and DROP?		
Ans	DELETE: removes selected rows, DML, can rollback. TRUNCATE: removes all rows, DDL, resets identity, cannot rollback. DROP: removes entire table structure, DDL.		
Co.	TCS • Amazon • Oracle • Microsoft	BL	L2 – Understand
Ex.	`DELETE = remove data. TRUNCATE = empty the table. DROP = destroy the table itself.`		

Q31	Difference between WHERE and HAVING?		
Ans	WHERE filters rows before grouping/aggregation. HAVING filters groups after aggregation (used with GROUP BY).		
Co.	TCS • Infosys • Amazon • Flipkart • Wipro	BL	L2 – Understand
Ex.	`SELECT dept_id, COUNT(*) FROM Employees WHERE salary>30000 GROUP BY dept_id HAVING COUNT(*)>3;`		

Q32	Difference between UNION and UNION ALL?		
Ans	UNION removes duplicate records from the combined result. UNION ALL keeps all records including duplicates (and is faster).		
Co.	TCS • Infosys • Oracle • Amazon	BL	L2 – Understand
Ex.	`SELECT city FROM Customers UNION SELECT city FROM Suppliers; — distinct cities only.`		

Q33	Difference between CHAR and VARCHAR?		
Ans	CHAR has a fixed length (padded with spaces if shorter); VARCHAR has variable length (only uses as much space as needed). CHAR is slightly faster; VARCHAR saves space.		
Co.	TCS • Infosys • Wipro • Oracle • HCL	BL	L2 – Understand
Ex.	`CHAR(10) for 'Hi' stores 'Hi        ' (8 spaces). VARCHAR(10) for 'Hi' stores just 'Hi'.`		

Q34	Difference between delete and drop?		
Ans	DELETE removes rows from a table (data only), the table still exists. DROP removes the entire table structure along with all data.		
Co.	TCS • Wipro • HCL	BL	L2 – Understand
Ex.	`After DELETE, the table is empty but still exists. After DROP, the table is completely gone.`		

Q35	What is a sequence in SQL?		
Ans	A database object that generates a sequential series of numeric values, commonly used for auto-generating primary key values.		
Co.	Oracle • IBM • SAP • PostgreSQL	BL	L2 – Understand
Ex.	`CREATE SEQUENCE emp_seq START WITH 1 INCREMENT BY 1; — then INSERT uses NEXTVAL('emp_seq') for the ID.`		

Q36	What is a savepoint in SQL?		
Ans	A marker within a transaction that allows partial rollback — you can undo to the savepoint without cancelling the entire transaction.		
Co.	Oracle • IBM • Microsoft	BL	L3 – Apply
Ex.	`SAVEPOINT sp1; INSERT ...; UPDATE ...; ROLLBACK TO SAVEPOINT sp1; COMMIT; — undoes only the insert/update after sp1.`		

Q37	What is the difference between COMMIT and ROLLBACK?		
Ans	COMMIT makes all changes in the current transaction permanent and visible. ROLLBACK undoes all changes since the last COMMIT, restoring the previous state.		
Co.	Oracle • IBM • TCS • Infosys	BL	L2 – Understand
Ex.	`BEGIN; UPDATE Employees SET salary=salary*1.1; ROLLBACK; — salary reverts. Use COMMIT to make it permanent.`		

Q38	What is a pivot query?		
Ans	A pivot query transforms row data into columns, used for data summarization and cross-tabulation reports.		
Co.	Microsoft • Oracle • Amazon • Flipkart	BL	L3 – Apply
Ex.	`Pivot months into columns: rows become month names, columns show sales figures per month per region.`		

Q39	What is ORM?		
Ans	Object-Relational Mapping — a technique to map database tables to programming language objects, allowing DB operations via code without raw SQL.		
Co.	Amazon • Flipkart • Zomato • Microsoft	BL	L2 – Understand
Ex.	`Hibernate (Java), SQLAlchemy (Python), Django ORM — save/update/delete records as objects: employee.save().`		

Q40	What is a data dictionary?		
Ans	A repository storing metadata about database objects — table names, column names, data types, constraints, indexes, and relationships.		
Co.	Oracle • IBM • SAP	BL	L1 – Remember
Ex.	`In Oracle, SELECT * FROM information_schema.columns; shows all column metadata in the data dictionary.`		

Q41	Difference between temporary table and table variable?		
Ans	Temporary table: persists for the session or stored procedure, can have indexes and statistics. Table variable: limited scope, usually faster for small datasets, cannot have non-clustered indexes.		
Co.	Microsoft • Oracle • Amazon	BL	L3 – Apply
Ex.	`CREATE TEMPORARY TABLE temp_emps AS SELECT * FROM Employees WHERE dept_id=3; — session-level temp table.`		

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Q42	What is UPSERT? How do you implement it?		
Ans	UPSERT inserts a new row if it doesn't exist, or updates it if it does. MySQL uses INSERT...ON DUPLICATE KEY UPDATE; PostgreSQL uses INSERT...ON CONFLICT DO UPDATE.		
Co.	Amazon • Google • Flipkart • Zomato	BL	L3 – Apply
Ex.	□ `INSERT INTO Employee(emp_id, salary) VALUES(101, 75000) ON CONFLICT(emp_id) DO UPDATE SET salary=EXCLUDED.salary;`		

4. Joins & Subqueries

Q43	What is a join in SQL?		
Ans	A JOIN combines rows from two or more tables based on a related column (usually a key relationship) to produce a unified result set.		
Co.	TCS • Infosys • Amazon • Flipkart • Google	BL	L1 – Remember
Ex.	□ `SELECT e.name, d.dept_name FROM Employees e JOIN Departments d ON e.dept_id = d.dept_id;`		

Q44	What are different types of joins?		
Ans	INNER JOIN (matching rows only), LEFT OUTER JOIN (all left + matching right), RIGHT OUTER JOIN (all right + matching left), FULL OUTER JOIN (all rows from both), SELF JOIN (table with itself), CROSS JOIN (cartesian product).		
Co.	TCS • Infosys • Amazon • Google • Microsoft	BL	L2 – Understand
Ex.	□ `LEFT JOIN: keeps all employees even if they have no department assigned.`		

Q45	What is a self-join?		
Ans	A join where a table is joined with itself — used to compare rows within the same table, often for hierarchical data.		
Co.	Amazon • Google • Microsoft • Zomato	BL	L3 – Apply
Ex.	□ `SELECT e.name, m.name AS manager FROM Employees e JOIN Employees m ON e.manager_id = m.emp_id;`		

Q46	What is a subquery?		
Ans	A query nested inside another SQL query. It can appear in SELECT, FROM, or WHERE clauses and returns data used by the outer query.		
Co.	TCS • Infosys • Amazon • Wipro	BL	L2 – Understand
Ex.	□ `SELECT name FROM Employees WHERE salary > (SELECT AVG(salary) FROM Employees);`		

Q47	What is a correlated subquery?		
Ans	A subquery that references columns from the outer query. It executes once for each row processed by the outer query — making it potentially slower.		
Co.	Amazon • Google • Oracle • Microsoft	BL	L4 – Analyse

Ex.	```SELECT e1.name FROM Employees e1 WHERE salary > (SELECT AVG(salary) FROM Employees e2 WHERE e2.dept_id = e1.dept_id);```

Q48	Difference between inner join and outer join?		
Ans	INNER JOIN returns only matching rows from both tables. OUTER JOIN (LEFT/RIGHT/FULL) returns all rows from one or both tables, with NULLs for non-matching rows.		
Co.	TCS • Infosys • Amazon • Oracle	BL	L2 – Understand
Ex.	INNER JOIN loses employees without departments. LEFT JOIN keeps all employees, NULL for dept if none.		

Q49	What is a union vs join?		
Ans	UNION combines rows vertically from two similar queries (stacking result sets). JOIN combines tables horizontally based on key relationships.		
Co.	TCS • Oracle • IBM	BL	L2 – Understand
Ex.	UNION: SELECT name FROM Students UNION SELECT name FROM Teachers — stacks names. JOIN: links tables by key.		

Q50	Difference between inner join and cross join?		
Ans	INNER JOIN returns rows where join condition is met. CROSS JOIN returns the Cartesian product — every row of table A paired with every row of table B (no condition).		
Co.	Amazon • Oracle • Microsoft	BL	L3 – Apply
Ex.	If A has 5 rows and B has 4 rows, CROSS JOIN returns 20 rows. INNER JOIN returns only matching rows.		

Q51	What is a hash join?		
Ans	A join algorithm that builds a hash table from the smaller table and probes it with rows from the larger table. Efficient for large datasets with no suitable index.		
Co.	Amazon • Google • Oracle • Microsoft	BL	L4 – Analyse
Ex.	Query optimizer chooses hash join when joining two large tables without indexes on join columns.		

Q52	What is a nested loop join?		
Ans	A join algorithm that compares each row of one table (outer) with every row of another table (inner). Simple but O(n×m) — slow for large datasets.		
Co.	Oracle • IBM • Amazon	BL	L4 – Analyse
Ex.	Best for small tables or when inner table has an index on the join column.		

Q53	Which is faster: EXISTS or IN?		
Ans	EXISTS is faster when the subquery returns many records (it stops on first match). IN is faster for small, indexed subquery result sets.		
Co.	Amazon • Oracle • Microsoft • TCS	BL	L4 – Analyse

Ex.	□ EXISTS: SELECT * FROM Employees e WHERE EXISTS(SELECT 1 FROM Dept d WHERE d.dept_id=e.dept_id AND d.location='NYC');

5. Indexes

Q54	What is an index?		
Ans	A data structure that speeds up data retrieval operations on a database table at the cost of additional storage space and slower write operations.		
Co.	TCS • Infosys • Amazon • Google • Oracle	BL	L1 – Remember
Ex.	□ CREATE INDEX idx_salary ON Employees(salary); — makes WHERE salary > 50000 queries much faster.		

Q55	What are indexes used for?		
Ans	To improve query performance and speed of data retrieval by avoiding full table scans. They create shortcuts to data using B-Tree or Hash structures.		
Co.	TCS • Amazon • Oracle • Microsoft	BL	L2 – Understand
Ex.	□ Without index: MySQL scans all 10M rows. With index: jumps directly to matching rows in milliseconds.		

Q56	What is a clustered index?		
Ans	Data rows in the table are physically stored in the order of the clustered index. Only one clustered index can exist per table.		
Co.	Amazon • Microsoft • Oracle • TCS	BL	L2 – Understand
Ex.	□ SQL Server: Primary Key creates a clustered index by default — data is physically sorted by emp_id.		

Q57	What is a non-clustered index?		
Ans	A separate structure from the data that stores the indexed columns and pointers (row locators) to the actual data rows. Multiple non-clustered indexes can exist.		
Co.	Amazon • Microsoft • Oracle • TCS	BL	L2 – Understand
Ex.	□ CREATE INDEX idx_name ON Employees(name); — creates a separate B-Tree with name→row pointer mapping.		

Q58	Difference between clustered and non-clustered index?		
Ans	Clustered: physical data stored in index order, one per table, faster for range queries. Non-clustered: separate structure with row pointers, multiple per table, slightly slower.		
Co.	Amazon • Microsoft • Oracle • TCS • Google	BL	L4 – Analyse
Ex.	□ Clustered index on emp_id: rows are in emp_id order on disk. Non-clustered on salary: separate lookup structure.		

Q59	Difference between clustered index and primary key?

Ans	Primary key is a logical constraint that enforces uniqueness. Clustered index is a physical storage concept. By default in SQL Server, PK creates a clustered index — but they are different concepts.		
Co.	Microsoft • Oracle • Amazon	BL	L4 – Analyse
Ex.	`□ You can have a clustered index on a non-PK column. PK without clustered index uses a heap table.`		

Q60	What is a covering index?		
Ans	An index that includes all the columns required by a query, so the query engine never needs to access the actual table rows — reducing I/O dramatically.		
Co.	Amazon • Google • Microsoft • Oracle	BL	L4 – Analyse
Ex.	`□ Query: SELECT name, salary FROM Employees WHERE dept_id=3. Index on (dept_id, name, salary) = covering index.`		

Q61	Explain types of indexes (beyond clustered/non-clustered).		
Ans	B-Tree/B+ Tree (balanced tree, supports range queries), Hash Index (equality searches only, O(1)), Bitmap Index (low-cardinality columns like gender/status — great for OLAP).		
Co.	Oracle • Amazon • Google • IBM	BL	L4 – Analyse
Ex.	`□ Bitmap index on gender='M'/'F': compact bitmaps make AND/OR operations extremely fast for OLAP queries.`		

Q62	What is a B-Tree index and why is it preferred in databases?		
Ans	A B-Tree (Balanced Tree) index keeps data sorted and supports searches, insertions, and deletions in O(log n). Preferred because it supports both equality and range queries and maps well to disk-based storage.		
Co.	Amazon • Google • Oracle • Microsoft • PostgreSQL	BL	L4 – Analyse
Ex.	`□ B-Tree on salary: WHERE salary BETWEEN 40000 AND 80000 uses the tree to find the range without scanning all rows.`		

Q63	What is a B+ Tree and how does it differ from a B-Tree?		
Ans	In a B+ Tree, all data pointers are stored only in leaf nodes (linked as a list). Internal nodes store only keys for navigation. This makes range queries very efficient. Most RDBMS (MySQL InnoDB, PostgreSQL) use B+ Trees.		
Co.	Amazon • Google • Oracle • MySQL	BL	L4 – Analyse
Ex.	`□ Range scan: B+ Tree leaf linked list lets the engine scan salary 40K-80K by traversing the leaf chain.`		

Q64	What is the difference between a hash index and a B-Tree index?		
Ans	Hash Index: uses a hash function, O(1) for equality lookups (=) only — no range queries. B-Tree Index: tree structure, O(log n), supports both equality and range queries (BETWEEN, LIKE prefix, >, <).		
Co.	Amazon • Google • Oracle • PostgreSQL	BL	L4 – Analyse
Ex.	`□ Hash: WHERE emp_id = 101 (single lookup). B-Tree: WHERE salary BETWEEN 40000 AND 80000 (range query).`		

Q65	What is a Bitmap Index?		
Ans	A Bitmap Index stores a bit array for each distinct value of a low-cardinality column. Very efficient for read-heavy OLAP queries; performs poorly with frequent INSERT/UPDATE/DELETE.		
Co.	Oracle • Amazon • Google	BL	L4 – Analyse
Ex.	`Bitmap index on gender: stores two bitmaps (M:1010011, F:0101100). AND/OR on multiple bitmaps is O(n/64).`		

Q66	What is a query execution plan? How do you analyze one?		
Ans	A step-by-step strategy the query optimizer chooses to execute SQL. Use EXPLAIN (MySQL/PostgreSQL) or EXPLAIN ANALYZE to view it. Look for full table scans, nested loops on large tables, and missing indexes as red flags.		
Co.	Amazon • Google • Microsoft • Oracle • Flipkart	BL	L5 – Evaluate
Ex.	`EXPLAIN ANALYZE SELECT * FROM Employees WHERE salary > 50000; — shows Seq Scan vs Index Scan.`		

Q67	What does Seq Scan vs Index Scan mean in a query execution plan?		
Ans	Seq Scan (Sequential Scan): reads every row — used when no suitable index exists or table is small. Index Scan: traverses index tree to find matching rows — faster for selective queries. Index Only Scan: satisfies query entirely from the index without touching the table.		
Co.	Amazon • Google • Flipkart • Zomato	BL	L5 – Evaluate
Ex.	`EXPLAIN on a million-row table: Seq Scan is bad; Index Scan on an indexed column is 100x faster.`		

6. Views & Stored Objects

Q68	What is a view?		
Ans	A virtual table created from one or more tables using a SELECT query. It does not store data physically — it is a saved query that behaves like a table.		
Co.	TCS • Infosys • Oracle • Amazon	BL	L2 – Understand
Ex.	`CREATE VIEW HighEarners AS SELECT name, salary FROM Employees WHERE salary > 80000; — used like: SELECT * FROM HighEarners;`		

Q69	What is a view in SQL?		
Ans	A virtual table derived from one or more tables using a SELECT statement. Views simplify complex queries, provide security by hiding sensitive columns, and present a consistent interface.		
Co.	TCS • Infosys • Oracle • IBM	BL	L2 – Understand
Ex.	`CREATE VIEW emp_dept AS SELECT e.name, d.dept_name FROM Employees e JOIN Departments d ON e.dept_id=d.dept_id;`		

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Q70	Difference between materialized view and view?		
Ans	A regular view is virtual (computed on every query). A materialized view is physically stored on disk and periodically refreshed — faster for complex queries but consumes storage.		
Co.	Oracle • Amazon • Google • PostgreSQL	BL	L4 – Analyse
Ex.	□ Materialized view for monthly sales report: stored once, refreshed nightly — no recalculation on each query.		

Q71	What is a stored procedure?		
Ans	A precompiled collection of SQL statements stored and executed on the database server. Reduces network traffic, improves performance, and encapsulates business logic.		
Co.	TCS • Infosys • Oracle • Microsoft • IBM	BL	L2 – Understand
Ex.	□ CREATE PROCEDURE GetEmployeeSalary(IN emp_id INT) BEGIN SELECT salary FROM Employees WHERE emp_id=emp_id; END;		

Q72	What is a trigger?		
Ans	A set of SQL statements automatically executed (fired) when a specified event (INSERT, UPDATE, DELETE) occurs on a table. Used for auditing, validation, and cascading changes.		
Co.	TCS • Infosys • Oracle • IBM • HCL	BL	L2 – Understand
Ex.	□ AFTER INSERT trigger on Orders automatically updates inventory count in the Products table.		

Q73	Difference between function and stored procedure?		
Ans	A function must return a value and can be used in SQL expressions; it generally cannot modify database state. A stored procedure may or may not return a value and can perform INSERT/UPDATE/DELETE operations.		
Co.	Oracle • Microsoft • TCS • Amazon	BL	L3 – Apply
Ex.	□ Function: SELECT getFullName(101); — returns a value. Procedure: EXEC transferFunds(101, 202, 5000); — modifies data.		

Q74	What is a cursor?		
Ans	A database object used to retrieve and process rows one by one from a result set. Cursors are useful when row-by-row processing is needed but are slower than set-based operations.		
Co.	Oracle • IBM • TCS • Microsoft	BL	L3 – Apply
Ex.	□ DECLARE cur CURSOR FOR SELECT emp_id FROM Employees; FETCH NEXT FROM cur INTO @id; — processes one row at a time.		

Q75	What is SQL Injection?		
Ans	A security attack where malicious SQL code is inserted into input fields to manipulate, read, or destroy database data. Prevented by parameterized queries and prepared statements.		
Co.	Amazon • Google • Microsoft • Flipkart • Zomato	BL	L5 – Evaluate
Ex.	□ Input: ' OR '1'='1 in login makes query: SELECT * FROM users WHERE user='' OR '1'='1' — returns all rows!		

7. Normalization & ER Modeling

Q76	What is normalization?		
Ans	A process of organizing data to reduce redundancy and improve data integrity by dividing large tables into smaller, related tables following normal form rules.		
Co.	TCS • Infosys • Wipro • Amazon • Oracle	BL	L2 – Understand
Ex.	□ Splitting Employees(emp_id, name, dept_name, dept_location) → Employees + Departments tables avoids duplicating dept_location.		

Q77	What are the normal forms?		
Ans	1NF: atomic (indivisible) values. 2NF: no partial dependency on part of composite key. 3NF: no transitive dependency (non-key → non-key). BCNF: every determinant is a super key.		
Co.	TCS • Infosys • Amazon • Oracle • Google	BL	L2 – Understand
Ex.	□ 1NF: No 'Physics,Math' in one cell. 2NF: Grade depends on full key (student_id, course_id), not just student_id.		

Q78	What is denormalization?		
Ans	Intentionally combining normalized tables to improve read/query performance, at the cost of data redundancy. Common in data warehouses and reporting systems.		
Co.	Amazon • Google • Microsoft • Oracle	BL	L3 – Apply
Ex.	□ Storing dept_name directly in Employees instead of joining — faster reads, but updating dept_name requires multiple rows.		

Q79	What is normalization important?		
Ans	Normalization eliminates data redundancy, prevents update/insert/delete anomalies, and ensures data consistency across the database.		
Co.	TCS • Infosys • Wipro	BL	L2 – Understand
Ex.	□ Without normalization: changing a manager's name requires updating hundreds of employee rows.		

Q80	What is Functional Dependency? How does it relate to normalization?		
Ans	A functional dependency X→Y means the value of X uniquely determines Y. Normalization removes partial FDs (2NF), transitive FDs (3NF), and FDs where determinant is not a super key (BCNF).		
Co.	Amazon • Oracle • Google • Microsoft	BL	L4 – Analyse
Ex.	□ In Student(student_id, course_id, student_name): student_id→student_name is a partial FD — violates 2NF.		

Q81	Explain BCNF with a simple example.		
Ans	BCNF requires that for every FD X→Y, X must be a super key. It is stronger than 3NF. A table can satisfy 3NF but still violate BCNF.		
Co.	Amazon • Google • Oracle • Microsoft	BL	L4 – Analyse

Ex.	□ Table: (Student, Subject, Teacher) where Teacher→Subject. Teacher is not super key → BCNF violation. Split into (Teacher, Subject) and (Student, Teacher).

Q82	What are 4NF and 5NF?		
Ans	4NF: eliminates multi-valued dependencies (a table should not contain two independent multi-valued facts about an entity). 5NF (Project-Join NF): eliminates join dependencies — table cannot be losslessly decomposed into smaller tables.		
Co.	Amazon • Google • Oracle	BL	L5 – Evaluate
Ex.	□ 4NF: Employee(emp_id, skill, hobby) — skill and hobby are independent MVDs. Split into Emp_Skill and Emp_Hobby.		

Q83	What are the different types of anomalies in an unnormalized database?		
Ans	Insertion anomaly: cannot insert data without inserting unrelated data. Deletion anomaly: deleting one record unintentionally removes unrelated data. Update anomaly: changing one value requires multiple updates, risking inconsistency.		
Co.	TCS • Oracle • IBM • Amazon	BL	L4 – Analyse
Ex.	□ Without normalization: deleting the last employee in a department also deletes the department's details (deletion anomaly).		

Q84	Difference between 3NF and BCNF with an example.		
Ans	3NF: no transitive dependency (minor exceptions allowed for lossless decomposition). BCNF: every determinant must be a candidate key — stricter. BCNF may not always preserve all functional dependencies.		
Co.	Amazon • Google • Oracle • Microsoft	BL	L5 – Evaluate
Ex.	□ (Student, Subject, Teacher): Teacher→Subject. This satisfies 3NF but violates BCNF since Teacher is not a candidate key.		

Q85	What is an ER model? Difference between ER model and relational model?		
Ans	ER model is a conceptual design showing entities, attributes, and relationships. Relational model is the implementation — entities become tables, relationships become foreign keys.		
Co.	TCS • Oracle • IBM • SAP	BL	L2 – Understand
Ex.	□ ER model shows Student-enrolls-in-Course. Relational model creates tables Student, Course, Enrollment with FKs.		

Q86	What are the different types of relationships in ER modeling?		
Ans	One-to-One (1:1): one entity relates to at most one other. One-to-Many (1:N): most common — e.g., one department has many employees. Many-to-Many (M:N): resolved with a junction table.		
Co.	TCS • Oracle • IBM • Amazon	BL	L2 – Understand
Ex.	□ M:N: Student and Course — a student takes many courses; a course has many students. Junction: Enrollment(student_id, course_id).		

Q87	What is a junction table and when do you use it?

Ans	A junction (bridge/associative) table resolves Many-to-Many relationships by holding the foreign keys of both related tables as a composite primary key.		
Co.	TCS • Amazon • Oracle • Flipkart	BL	L3 – Apply
Ex.	`Student_Course(student_id, course_id) links Students and Courses. It can also hold extra attributes like enrollment_date.`		

Q88	**Explain cardinality and participation constraints in the ER Model.**		
Ans	Cardinality: defines relationship type — 1:1, 1:N, M:N. Participation: Total (double line — every entity must participate) vs Partial (single line — entity may or may not participate).		
Co.	Amazon • Oracle • Google • IBM	BL	L4 – Analyse
Ex.	`Employee MUST belong to a Department (total participation). Department MAY have a manager (partial participation).`		

Q89	**What is the difference between conceptual, logical, and physical schema?**		
Ans	Conceptual: high-level ER diagram, technology-agnostic. Logical: tables, columns, data types, constraints — independent of DBMS. Physical: storage engine, indexes, partitions — DBMS-specific.		
Co.	Oracle • IBM • SAP • Amazon	BL	L3 – Apply
Ex.	`Conceptual: Student-enrolls-Course. Logical: Student(id INT, name VARCHAR). Physical: InnoDB engine, clustered index on id.`		

8. Transactions & Concurrency Control

Q90	**What is a transaction?**		
Ans	A logical unit of work containing one or more SQL operations that must all succeed (COMMIT) or all be undone (ROLLBACK) — treated as a single atomic operation.		
Co.	TCS • Infosys • Amazon • Oracle • IBM	BL	L2 – Understand
Ex.	`Bank transfer: Debit account A AND Credit account B. If credit fails, debit must also roll back.`		

Q91	**What are ACID properties?**		
Ans	Atomicity (all or nothing), Consistency (DB moves from one valid state to another), Isolation (concurrent transactions don't interfere), Durability (committed changes survive failures).		
Co.	TCS • Infosys • Amazon • Google • Oracle • Microsoft	BL	L2 – Understand
Ex.	`ATM transaction: Atomicity ensures money debited = money credited. Durability ensures committed balance survives power outage.`		

Q92	**What is a deadlock?**		
Ans	A situation where two or more transactions are each waiting for a lock held by the other, causing all to wait indefinitely. Resolved by killing one transaction (victim selection).		
Co.	Amazon • Google • Oracle • Microsoft • TCS	BL	L3 – Apply

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Ex.	T1 holds lock on A, wants B. T2 holds lock on B, wants A. Both wait forever = deadlock.

Q93	What is a transaction log?		
Ans	A record of all changes made to the database (before and after images), used for recovery (replaying committed transactions) and rollback of uncommitted transactions.		
Co.	Oracle • IBM • Microsoft • Amazon	BL	L3 – Apply
Ex.	After a server crash, the DB replays the transaction log to restore all committed changes and undo uncommitted ones.		

Q94	What is concurrency control in DBMS?		
Ans	A mechanism that manages simultaneous operations on the database, ensuring data consistency when multiple transactions execute in parallel without conflicting.		
Co.	Amazon • Google • Oracle • Microsoft	BL	L3 – Apply
Ex.	Two users booking the last airplane seat simultaneously — concurrency control ensures only one succeeds.		

Q95	What is a lock in DBMS? What are the types?		
Ans	A mechanism to control concurrent access. Shared (S-lock): allows concurrent reads, no writes. Exclusive (X-lock): allows read+write by one transaction only, blocks all others.		
Co.	Amazon • Oracle • Microsoft • Google	BL	L3 – Apply
Ex.	Reading employee record: S-lock (others can also read). Updating salary: X-lock (others must wait).		

Q96	What is Two-Phase Locking (2PL)?		
Ans	A concurrency protocol with Growing Phase (acquire locks, never release) and Shrinking Phase (release locks, never acquire new). Guarantees serializability but can cause deadlocks.		
Co.	Amazon • Oracle • Google • IBM	BL	L5 – Evaluate
Ex.	T1 acquires locks for all needed resources (growing), performs work, then releases all (shrinking) — no new locks after first release.		

Q97	What is a schedule in DBMS? Differentiate serial vs. serializable.		
Ans	A schedule is the ordering of operations from multiple concurrent transactions. Serial: one transaction runs to completion before the next. Serializable: concurrent schedule with outcome equivalent to some serial schedule.		
Co.	Amazon • Oracle • Google	BL	L4 – Analyse
Ex.	Serial: T1 fully completes, then T2 starts. Serializable: T1 and T2 interleave but produce same result as T1→T2 or T2→T1.		

Q98	What are transaction isolation levels?
Ans	Read Uncommitted (dirty reads allowed), Read Committed (prevents dirty reads), Repeatable Read (prevents dirty + non-repeatable reads), Serializable (strictest — prevents phantom reads too).

Co.	Amazon • Oracle • Microsoft • Google	BL	L4 – Analyse
Ex.	□ Banking: use Serializable. Analytics dashboard: Read Committed is fine. Real-time feed: Read Uncommitted may be acceptable.		

Q99	**What are phantom read, dirty read, and non-repeatable read?**		
Ans	Dirty read: reading uncommitted data from another transaction. Non-repeatable read: same row read twice gives different values (another committed update in between). Phantom read: re-executing a range query returns new rows (another committed insert).		
Co.	Amazon • Google • Oracle • Microsoft	BL	L4 – Analyse
Ex.	□ Dirty read: T1 reads T2's in-progress salary update, then T2 rolls back. T1 read invalid data.		

Q100	**What is a phantom read? How is it different from a dirty read?**		
Ans	Dirty read involves modified rows from uncommitted transactions. Phantom read involves new or missing rows — a transaction re-executes a range query and finds rows that weren't there before.		
Co.	Amazon • Google • Oracle	BL	L4 – Analyse
Ex.	□ T1: SELECT * WHERE age>30 → 10 rows. T2 inserts a person aged 35 and commits. T1 re-queries → 11 rows = phantom.		

Q101	**What is MVCC (Multi-Version Concurrency Control)?**		
Ans	A technique where the DB keeps multiple versions of each data item. Readers access an older consistent snapshot while writers create new versions — reads and writes don't block each other.		
Co.	Amazon • Google • PostgreSQL • MySQL	BL	L5 – Evaluate
Ex.	□ PostgreSQL: a SELECT sees data as it was at transaction start, even if another transaction is updating simultaneously.		

Q102	**What is WAL (Write-Ahead Logging)?**		
Ans	A recovery technique where all changes are written to a log file on disk before being applied to the actual database files. After a crash, the DB replays or undoes operations from the log.		
Co.	Amazon • Oracle • PostgreSQL • Google	BL	L5 – Evaluate
Ex.	□ PostgreSQL uses WAL: every INSERT/UPDATE writes to WAL first. On crash restart, WAL is replayed to restore consistency.		

Q103	**What are the different types of database failures?**		
Ans	Transaction Failure (logic error, user abort). System Failure (power loss/OS crash — buffer lost, disk OK). Media Failure (disk crash — persistent data lost, recovered from backup + logs).		
Co.	Oracle • IBM • Amazon • Microsoft	BL	L4 – Analyse
Ex.	□ System failure: buffer pool data lost → WAL replay restores committed transactions. Media failure: restore from last backup + replay WAL.		

9. Advanced SQL – Window Functions & CTEs

Q104	What are window functions in SQL? Name key ones.		
Ans	Window functions perform calculations across a set of rows related to the current row without collapsing rows like GROUP BY. Key functions: ROW_NUMBER(), RANK(), DENSE_RANK(), NTILE(), LAG(), LEAD(), FIRST_VALUE(), LAST_VALUE(), SUM() OVER(), AVG() OVER().		
Co.	Amazon • Google • Microsoft • Flipkart • Zomato	BL	L4 – Analyse
Ex.	`SELECT name, salary, RANK() OVER(ORDER BY salary DESC) AS rank FROM Employees; — ranks all employees by salary.`		

Q105	What is the difference between RANK() and DENSE_RANK()?		
Ans	RANK() leaves gaps after tied ranks (1,2,2,4). DENSE_RANK() does not leave gaps (1,2,2,3). Use DENSE_RANK for Nth distinct value; use RANK for exact positional standing.		
Co.	Amazon • Google • Flipkart • Zomato • Microsoft	BL	L4 – Analyse
Ex.	`Salaries: 90k, 80k, 80k, 70k. RANK: 1,2,2,4. DENSE_RANK: 1,2,2,3. Second highest = salary where DENSE_RANK=2.`		

Q106	What is LAG() and LEAD() in SQL?		
Ans	LAG(col, n) accesses the value of a column n rows before the current row. LEAD(col, n) accesses n rows ahead. Used for month-over-month comparisons and trend analysis.		
Co.	Amazon • Google • Flipkart • Microsoft • Zomato	BL	L4 – Analyse
Ex.	`SELECT emp_id, salary, LAG(salary) OVER(PARTITION BY dept_id ORDER BY join_date) AS prev_salary FROM Employee;`		

Q107	What is a CTE (Common Table Expression)? When would you use it over a subquery?		
Ans	A CTE is a named temporary result set defined with the WITH clause. Use over subqueries when: the same subquery is referenced multiple times, recursive queries are needed, or readability is important.		
Co.	Amazon • Google • Microsoft • Flipkart	BL	L4 – Analyse
Ex.	`WITH dept_avg AS (SELECT dept_id, AVG(salary) AS avg_sal FROM Employees GROUP BY dept_id) SELECT e.name FROM Employees e JOIN dept_avg d ON e.dept_id=d.dept_id WHERE e.salary > d.avg_sal;`		

Q108	What is CTE (Common Table Expression) in SQL? Difference from subquery?		
Ans	A CTE (WITH clause) is a temporary named result set — more readable and can be recursive. A subquery is nested and evaluated inline. CTEs improve readability and can reference themselves.		
Co.	Amazon • Google • Microsoft • Oracle • Flipkart	BL	L4 – Analyse
Ex.	`Complex hierarchical query: CTE is far more readable than deeply nested subqueries.`		

Q109	Write SQL using CTE to find the nth highest salary (parameterized).

Ans	Use DENSE_RANK() in a CTE, then filter by rank = n. Replace n with the desired rank.		
Co.	Amazon • Google • Microsoft • Adobe • Flipkart	BL	L6 – Create
Ex.	`WITH Ranked AS (SELECT Salary, DENSE_RANK() OVER(ORDER BY Salary DESC) AS rnk FROM Employee) SELECT Salary FROM Ranked WHERE rnk = 3; -- 3rd highest`		

Q110	What is a recursive CTE? Give a use case.		
Ans	A recursive CTE references itself with UNION ALL to handle hierarchical or tree-structured data. Use cases: organizational hierarchy (employee→manager), bill of materials, graph traversal.		
Co.	Amazon • Google • Microsoft • Oracle	BL	L6 – Create
Ex.	`WITH RECURSIVE Hierarchy AS (SELECT emp_id, name, manager_id, 1 AS level FROM Employee WHERE manager_id IS NULL UNION ALL SELECT e.emp_id, e.name, e.manager_id, h.level+1 FROM Employee e JOIN Hierarchy h ON e.manager_id=h.emp_id) SELECT * FROM Hierarchy;`		

Q111	How do you find the running total (cumulative sum) in SQL?		
Ans	Use SUM() as a window function with ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW to accumulate totals row by row.		
Co.	Amazon • Google • Flipkart • Zomato • Microsoft	BL	L6 – Create
Ex.	`SELECT emp_id, salary, SUM(salary) OVER(PARTITION BY dept_id ORDER BY emp_id ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW) AS running_total FROM Employee;`		

Q112	How do you find gaps in a sequence of IDs?		
Ans	LEFT JOIN the table against a generated sequence of all IDs between min and max, then filter where the join found no match.		
Co.	Amazon • Google • Microsoft • Flipkart	BL	L6 – Create
Ex.	`SELECT s.id AS missing_id FROM (SELECT generate_series(MIN(id), MAX(id)) AS id FROM Employee) s LEFT JOIN Employee e ON s.id=e.id WHERE e.id IS NULL;`		

Q113	How do you swap two column values in a single UPDATE?		
Ans	In MySQL/PostgreSQL you can swap values in a single UPDATE without a temporary variable by assigning both columns simultaneously.		
Co.	Amazon • Google • Oracle • Microsoft	BL	L6 – Create
Ex.	`UPDATE Employee SET first_name = last_name, last_name = first_name WHERE emp_id = 1;`		

Q114	How do you calculate year-over-year (YoY) growth using SQL?		
Ans	Use LAG() partitioned by group to access the previous year's value, then compute percentage change using (current - previous) / previous * 100.		
Co.	Amazon • Google • Flipkart • Zomato • Microsoft	BL	L6 – Create
Ex.	`SELECT dept_id, yr, total_salary, LAG(total_salary) OVER(PARTITION BY dept_id ORDER BY yr) AS prev_year, ROUND((total_salary - LAG(total_salary) OVER(PARTITION BY dept_id ORDER BY yr))*100.0 / LAG(total_salary)`		

	OVER(PARTITION BY dept_id ORDER BY yr),2) AS yoy_pct FROM (SELECT dept_id, YEAR(join_date) AS yr, SUM(salary) AS total_salary FROM Employee GROUP BY dept_id, YEAR(join_date)) t;

10. SQL Query Practice Problems

Q115	**Given Employee(emp_id, emp_name, dept_id, salary), find the second highest salary.**		
Ans	Use DENSE_RANK() window function or subquery to find the salary with rank = 2.		
Co.	Amazon • Google • Microsoft • Flipkart • Zomato • Adobe	BL	L6 – Create
Ex.	SELECT salary FROM (SELECT salary, DENSE_RANK() OVER(ORDER BY salary DESC) AS rnk FROM Employee) t WHERE rnk=2;		

Q116	**Find employees who earn more than the average salary.**		
Ans	Use a subquery to compute AVG(salary) and compare each employee's salary against it.		
Co.	TCS • Infosys • Amazon • Wipro • Cognizant	BL	L4 – Analyse
Ex.	SELECT name, salary FROM Employees WHERE salary > (SELECT AVG(salary) FROM Employees);		

Q117	**Count employees in each department.**		
Ans	Use GROUP BY on dept_id with COUNT(*) aggregate function.		
Co.	TCS • Infosys • Wipro • HCL • Accenture	BL	L3 – Apply
Ex.	SELECT dept_id, COUNT(*) AS emp_count FROM Employees GROUP BY dept_id ORDER BY emp_count DESC;		

Q118	**Find departments with more than 5 employees.**		
Ans	Use GROUP BY + HAVING COUNT(*) > 5.		
Co.	TCS • Amazon • Flipkart • Zomato	BL	L3 – Apply
Ex.	SELECT dept_id, COUNT(*) FROM Employees GROUP BY dept_id HAVING COUNT(*) > 5;		

Q119	**Find employees who are managers (appear as manager_id in Employee table).**		
Ans	Use a subquery with IN or EXISTS to find emp_ids that appear as manager_id values.		
Co.	Amazon • Google • Microsoft	BL	L4 – Analyse
Ex.	SELECT DISTINCT e.name FROM Employees e WHERE e.emp_id IN (SELECT DISTINCT manager_id FROM Employees WHERE manager_id IS NOT NULL);		

Q120	**Retrieve employees whose names start with 'A'.**		
Ans	Use LIKE 'A%' pattern matching in the WHERE clause.		
Co.	TCS • Infosys • Wipro	BL	L2 – Understand
Ex.	SELECT * FROM Employees WHERE name LIKE 'A%';		

Q121	Find employees with no manager.		
Ans	Filter rows where manager_id IS NULL.		
Co.	TCS • Amazon • Flipkart	BL	L2 – Understand
Ex.	`SELECT name FROM Employees WHERE manager_id IS NULL;`		

Q122	Delete duplicate rows keeping the latest emp_id.		
Ans	Use a CTE with ROW_NUMBER() to identify duplicates, then delete older rows.		
Co.	Amazon • Google • Oracle	BL	L6 – Create
Ex.	`WITH dups AS (SELECT emp_id, ROW_NUMBER() OVER(PARTITION BY name, dept_id ORDER BY emp_id DESC) AS rn FROM Employees) DELETE FROM Employees WHERE emp_id IN (SELECT emp_id FROM dups WHERE rn > 1);`		

Q123	Find employees earning the maximum salary per department.		
Ans	Use MAX() as a window function or subquery grouped by dept_id.		
Co.	Amazon • Google • Flipkart • Zomato	BL	L4 – Analyse
Ex.	`SELECT name, dept_id, salary FROM Employees WHERE (dept_id, salary) IN (SELECT dept_id, MAX(salary) FROM Employees GROUP BY dept_id);`		

Q124	Find all departments that have employees earning above 50k.		
Ans	Join with Departments, filter salary > 50000, and return distinct department names.		
Co.	TCS • Amazon • Wipro	BL	L3 – Apply
Ex.	`SELECT DISTINCT d.dept_name FROM Departments d JOIN Employees e ON d.dept_id=e.dept_id WHERE e.salary > 50000;`		

Q125	Find employees who earn more than their manager.		
Ans	Self-join Employees table — compare employee's salary with their manager's salary.		
Co.	Amazon • Google • Microsoft • Adobe	BL	L5 – Evaluate
Ex.	`SELECT e.name FROM Employees e JOIN Employees m ON e.manager_id=m.emp_id WHERE e.salary > m.salary;`		

Q126	Find the third highest salary without using LIMIT or ROWNUM.		
Ans	Use a correlated subquery that counts how many distinct salaries are greater than the current row's salary, and filter where that count equals 2.		
Co.	Amazon • Google • Microsoft	BL	L6 – Create
Ex.	`SELECT DISTINCT salary FROM Employees e WHERE 2=(SELECT COUNT(DISTINCT salary) FROM Employees WHERE salary > e.salary);`		

Q127	Find consecutive employees with the same salary.
Ans	Use LAG() to compare each row's salary with the previous row's salary.

Co.	Amazon • Google • Flipkart	BL	L5 – Evaluate
Ex.	SELECT emp_id, salary FROM (SELECT emp_id, salary, LAG(salary) OVER(ORDER BY emp_id) AS prev_sal FROM Employees) t WHERE salary = prev_sal;		

Q128	Count the number of employees who joined in the last 6 months.		
Ans	Use DATEADD/INTERVAL with GETDATE()/NOW() to filter by recent join dates.		
Co.	TCS • Amazon • Wipro	BL	L3 – Apply
Ex.	SELECT COUNT(*) FROM Employees WHERE join_date >= DATEADD(MONTH, -6, GETDATE());		

Q129	Display the salary rank of each employee.		
Ans	Use RANK() or DENSE_RANK() as a window function ordered by salary DESC.		
Co.	Amazon • Flipkart • Zomato	BL	L3 – Apply
Ex.	SELECT name, salary, DENSE_RANK() OVER(ORDER BY salary DESC) AS salary_rank FROM Employees;		

Q130	Find employees who are not assigned to any project.		
Ans	Use LEFT JOIN with Projects table and filter WHERE project_id IS NULL.		
Co.	TCS • Amazon • Wipro • Cognizant	BL	L4 – Analyse
Ex.	SELECT e.name FROM Employees e LEFT JOIN Employee_Projects ep ON e.emp_id=ep.emp_id WHERE ep.project_id IS NULL;		

Q131	Find the second lowest salary using DENSE_RANK().		
Ans	Use DENSE_RANK() ordered by salary ASC and filter where rank = 2.		
Co.	Amazon • Google • Flipkart	BL	L4 – Analyse
Ex.	SELECT salary FROM (SELECT salary, DENSE_RANK() OVER(ORDER BY salary ASC) AS rnk FROM Employees) t WHERE rnk=2;		

Q132	Find employees working in all departments.		
Ans	Compare the count of distinct departments for each employee against the total department count.		
Co.	Amazon • Google • Oracle	BL	L6 – Create
Ex.	SELECT emp_id FROM Employee_Dept GROUP BY emp_id HAVING COUNT(DISTINCT dept_id) = (SELECT COUNT(*) FROM Departments);		

Q133	Find employees with salary above average in their department.		
Ans	Use a correlated subquery or window function (AVG OVER PARTITION BY dept_id) to compare.		
Co.	Amazon • Google • Flipkart • Microsoft	BL	L5 – Evaluate
Ex.	SELECT name, salary, dept_id FROM Employees e WHERE salary > (SELECT AVG(salary) FROM Employees WHERE dept_id = e.dept_id);		

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Q134	Pivot table: Show total salary per department per month.		
Ans	Use conditional aggregation (CASE WHEN) or database-specific PIVOT syntax to transform months into columns.		
Co.	Amazon • Microsoft • Oracle	BL	L6 – Create
Ex.	`SELECT dept_id, SUM(CASE WHEN MONTH(join_date)=1 THEN salary ELSE 0 END) AS Jan, SUM(CASE WHEN MONTH(join_date)=2 THEN salary ELSE 0 END) AS Feb FROM Employees GROUP BY dept_id;`		

Q135	Find employees whose name is a palindrome.		
Ans	Compare the name with its reverse using REVERSE() function.		
Co.	Amazon • Google • Microsoft	BL	L5 – Evaluate
Ex.	`SELECT name FROM Employees WHERE name = REVERSE(name);`		

Q136	Find departments with highest total salary.		
Ans	GROUP BY dept_id, sum salaries, order DESC, and take the top row.		
Co.	TCS • Amazon • Flipkart	BL	L3 – Apply
Ex.	`SELECT dept_id, SUM(salary) AS total FROM Employees GROUP BY dept_id ORDER BY total DESC LIMIT 1;`		

Q137	Find employees who joined on the same day as their manager.		
Ans	Self-join Employees on manager_id and compare join_date.		
Co.	Amazon • Google	BL	L5 – Evaluate
Ex.	`SELECT e.name FROM Employees e JOIN Employees m ON e.manager_id=m.emp_id WHERE DATE(e.join_date)=DATE(m.join_date);`		

Q138	Find the longest consecutive sequence of salaries.		
Ans	Use the island-gaps technique with ROW_NUMBER() to group consecutive values.		
Co.	Amazon • Google • Microsoft	BL	L6 – Create
Ex.	`Complex: group consecutive salary sequences using ROW_NUMBER() - salary difference trick to identify islands.`		

Q139	Find employees whose salary is unique.		
Ans	Group by salary, having COUNT = 1.		
Co.	TCS • Amazon • Wipro	BL	L3 – Apply
Ex.	`SELECT salary FROM Employees GROUP BY salary HAVING COUNT(*) = 1;`		

Q140	Delete all employees who have never managed anyone.		
Ans	DELETE WHERE emp_id NOT IN (SELECT DISTINCT manager_id FROM Employees WHERE manager_id IS NOT NULL).		
Co.	Amazon • Oracle • Microsoft	BL	L5 – Evaluate

Ex.	□ DELETE FROM Employees WHERE emp_id NOT IN (SELECT DISTINCT manager_id FROM Employees WHERE manager_id IS NOT NULL);

Q141	Find duplicate employee names along with count.		
Ans	GROUP BY name with HAVING COUNT(*) > 1.		
Co.	TCS • Amazon • Wipro • Flipkart	BL	L3 – Apply
Ex.	□ SELECT name, COUNT(*) AS cnt FROM Employees GROUP BY name HAVING COUNT(*) > 1;		

Q142	Retrieve employees whose names contain all vowels.		
Ans	Use multiple LIKE conditions — one for each vowel (a, e, i, o, u).		
Co.	Amazon • Google • Microsoft	BL	L4 – Analyse
Ex.	□ SELECT name FROM Employees WHERE LOWER(name) LIKE '%a%' AND LOWER(name) LIKE '%e%' AND LOWER(name) LIKE '%i%' AND LOWER(name) LIKE '%o%' AND LOWER(name) LIKE '%u%';		

Q143	Find departments with no employees.		
Ans	LEFT JOIN Employees onto Departments and filter WHERE emp_id IS NULL.		
Co.	TCS • Amazon • Wipro	BL	L3 – Apply
Ex.	□ SELECT d.dept_name FROM Departments d LEFT JOIN Employees e ON d.dept_id=e.dept_id WHERE e.emp_id IS NULL;		

Q144	Display the cumulative salary per department.		
Ans	Use SUM() OVER(PARTITION BY dept_id ORDER BY emp_id ROWS UNBOUNDED PRECEDING).		
Co.	Amazon • Google • Flipkart • Zomato	BL	L5 – Evaluate
Ex.	□ SELECT emp_id, dept_id, salary, SUM(salary) OVER(PARTITION BY dept_id ORDER BY emp_id ROWS BETWEEN UNBOUNDED PRECEDING AND CURRENT ROW) AS cum_salary FROM Employees;		

Q145	Find employees whose salary is above 90th percentile.		
Ans	Use PERCENTILE_CONT(0.90) or compare with a subquery using NTILE(10).		
Co.	Amazon • Google • Microsoft	BL	L6 – Create
Ex.	□ SELECT name, salary FROM Employees WHERE salary > (SELECT PERCENTILE_CONT(0.90) WITHIN GROUP(ORDER BY salary) FROM Employees);		

Q146	Find employees whose names are anagrams of each other.		
Ans	Sort the characters of each name and compare — names with same sorted characters are anagrams.		
Co.	Amazon • Google	BL	L6 – Create
Ex.	□ Complex: requires user-defined function to sort chars, then GROUP BY sorted_name HAVING COUNT > 1.		

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Q147	Find employees with salary increase trend month over month.		
Ans	Use LAG() with PARTITION BY emp_id ORDER BY month to compare consecutive salaries.		
Co.	Amazon • Google • Flipkart • Zomato	BL	L6 – Create
Ex.	`SELECT emp_id FROM salary_history WHERE salary > LAG(salary) OVER(PARTITION BY emp_id ORDER BY month);`		

Q148	Find employees who have the same salary as at least one other employee.		
Ans	JOIN the table with itself on salary, ensure emp_ids are different.		
Co.	Amazon • Google • Flipkart	BL	L4 – Analyse
Ex.	`SELECT DISTINCT e1.name FROM Employees e1 JOIN Employees e2 ON e1.salary=e2.salary AND e1.emp_id <> e2.emp_id;`		

Q149	Delete duplicate rows but keep one.		
Ans	Use CTE with ROW_NUMBER() to keep row with rn=1 and delete the rest.		
Co.	Amazon • Oracle • Microsoft	BL	L6 – Create
Ex.	`WITH cte AS (SELECT *, ROW_NUMBER() OVER(PARTITION BY name, salary ORDER BY emp_id) AS rn FROM Employees) DELETE FROM Employees WHERE emp_id IN (SELECT emp_id FROM cte WHERE rn > 1);`		

Q150	Find the manager with maximum employees.		
Ans	GROUP BY manager_id, count employees, order by count DESC, limit 1.		
Co.	Amazon • Google • Flipkart	BL	L4 – Analyse
Ex.	`SELECT ManagerID, COUNT(*) AS cnt FROM Employees GROUP BY ManagerID ORDER BY cnt DESC LIMIT 1;`		

Q151	Find consecutive days of attendance.		
Ans	Use the island-gaps technique: subtract ROW_NUMBER from AttendanceDate to get groups of consecutive days.		
Co.	Amazon • Google • Zomato	BL	L6 – Create
Ex.	`SELECT EmpID, MIN(AttendanceDate) AS Start, MAX(AttendanceDate) AS End FROM (SELECT EmpID, AttendanceDate, DATEADD(DAY,-ROW_NUMBER() OVER(PARTITION BY EmpID ORDER BY AttendanceDate), AttendanceDate) AS grp FROM Attendance) t GROUP BY EmpID, grp;`		

Q152	Rank employees by salary without using RANK/DENSE_RANK.		
Ans	Self-join: count how many employees have higher salary, add 1 to get the rank.		
Co.	Amazon • Google • Microsoft • Adobe	BL	L6 – Create
Ex.	`SELECT e1.Name, 1+COUNT(e2.Salary) AS Rank FROM Employees e1 LEFT JOIN Employees e2 ON e1.Salary < e2.Salary GROUP BY e1.Name, e1.Salary ORDER BY Rank;`		

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Q153	Find departments with average salary above 50k.		
Ans	Join Employees and Departments, GROUP BY department, HAVING AVG(salary) > 50000.		
Co.	TCS • Amazon • Flipkart • Zomato	BL	L3 – Apply
Ex.	SELECT DeptName FROM Departments d JOIN Employees e ON d.DeptID=e.DeptID GROUP BY DeptName HAVING AVG(Salary) > 50000;		

Q154	Find employees who never got a bonus.		
Ans	LEFT JOIN Employees with Bonus table and filter WHERE Bonus.emp_id IS NULL.		
Co.	TCS • Amazon • Wipro	BL	L4 – Analyse
Ex.	SELECT e.Name FROM Employees e LEFT JOIN Bonus b ON e.emp_id=b.emp_id WHERE b.emp_id IS NULL;		

Q155	Get department-wise salary percentage of total salary.		
Ans	Divide dept salary sum by total salary sum (subquery) and multiply by 100.		
Co.	Amazon • Google • Microsoft	BL	L5 – Evaluate
Ex.	SELECT DeptID, SUM(Salary)*100.0/(SELECT SUM(Salary) FROM Employees) AS SalaryPct FROM Employees GROUP BY DeptID;		

Q156	Find employees with salary higher than at least two colleagues.		
Ans	Use a correlated subquery: count employees with salary less than current, filter >= 2.		
Co.	Amazon • Google • Flipkart	BL	L5 – Evaluate
Ex.	SELECT e1.Name FROM Employees e1 WHERE 2 <= (SELECT COUNT(*) FROM Employees e2 WHERE e2.Salary < e1.Salary);		

Q157	Find employees whose name contains all vowels.		
Ans	Use multiple LIKE conditions for each vowel (case-insensitive).		
Co.	Amazon • Google • Microsoft	BL	L4 – Analyse
Ex.	SELECT Name FROM Employees WHERE Name LIKE '%a%' AND Name LIKE '%e%' AND Name LIKE '%i%' AND Name LIKE '%o%' AND Name LIKE '%u%';		

Q158	Find the difference between max and min salary in each department.		
Ans	GROUP BY dept_id, use MAX(salary) - MIN(salary).		
Co.	TCS • Amazon • Wipro • Flipkart	BL	L3 – Apply
Ex.	SELECT DeptID, MAX(Salary)-MIN(Salary) AS SalaryDiff FROM Employees GROUP BY DeptID;		

Q159	Find employees whose salary is greater than the average but less than the maximum salary of their department.		
Ans	Use two correlated subqueries: one for global AVG and one for dept-wise MAX.		
Co.	Amazon • Google • Microsoft • Adobe	BL	L6 – Create

Ex.	SELECT Name, Salary FROM Employees e WHERE Salary>(SELECT AVG(Salary) FROM Employees) AND Salary<(SELECT MAX(Salary) FROM Employees WHERE DeptID=e.DeptID);

11. Data Warehousing & NoSQL

Q160	What is a data warehouse?		
Ans	A centralized repository for storing large volumes of structured data from multiple operational sources, optimized for analytical queries and reporting.		
Co.	Amazon • Google • Microsoft • Oracle • SAP	BL	L2 – Understand
Ex.	Amazon Redshift, Google BigQuery, Snowflake — stores historical sales, clickstream data for business intelligence.		

Q161	Difference between OLTP and OLAP?		
Ans	OLTP (Online Transaction Processing): handles frequent small transactions (INSERT/UPDATE/DELETE), normalized schema. OLAP (Online Analytical Processing): handles complex analytical queries on large historical datasets, denormalized.		
Co.	Amazon • Oracle • SAP • IBM • Microsoft	BL	L3 – Apply
Ex.	OLTP: bank ATM transactions. OLAP: quarterly sales analysis across all regions and product lines.		

Q162	What is a star schema?		
Ans	A data warehouse schema with a central fact table (containing measurable data) surrounded by dimension tables. Simple structure, faster analytical queries.		
Co.	Amazon • Oracle • SAP • Microsoft	BL	L3 – Apply
Ex.	Fact_Sales(sale_id, date_id, product_id, customer_id, amount) connected to Dim_Date, Dim_Product, Dim_Customer.		

Q163	What is a snowflake schema?		
Ans	A normalized version of the star schema where dimension tables are further split into sub-dimension tables, reducing redundancy but increasing join complexity.		
Co.	Amazon • Oracle • SAP	BL	L3 – Apply
Ex.	Dim_Product → Dim_Category → Dim_Department. More normalized than star schema but requires more joins.		

Q164	What is NoSQL? When to choose over RDBMS?		
Ans	Non-relational databases for unstructured or semi-structured data. Choose NoSQL for high scalability, horizontal scaling, high-velocity/variety data. Use RDBMS for strict ACID compliance and structured relational data.		
Co.	Amazon • Google • Microsoft • MongoDB • Redis	BL	L4 – Analyse
Ex.	Twitter timeline = NoSQL (high write speed, flexible schema). Bank ledger = RDBMS (strict ACID required).		

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Q165	What are the types of NoSQL databases?		
Ans	Key-Value (Redis, DynamoDB — fast lookups). Document (MongoDB, CouchDB — JSON documents). Column-Family (Cassandra, HBase — column-based). Graph (Neo4j — nodes and edges, social networks).		
Co.	Amazon • Google • Microsoft • MongoDB	BL	L3 – Apply
Ex.	User sessions → Redis (key-value). Product catalog → MongoDB (document). Social network → Neo4j (graph).		

Q166	What is sharding in databases?		
Ans	Horizontal partitioning of data across multiple database instances (shards), each holding a subset. Improves write scalability and handles datasets larger than a single machine.		
Co.	Amazon • Google • Microsoft • Flipkart	BL	L4 – Analyse
Ex.	User data sharded by user_id: users 1-1M on shard1, 1M-2M on shard2. Distributes load across servers.		

Q167	What is replication in databases? Types?		
Ans	Copying data from one node (primary) to others (replicas). Synchronous: replica confirms write before primary acknowledges (strong consistency, slower). Asynchronous: primary acknowledges immediately, replica catches up later (faster, possible data lag).		
Co.	Amazon • Google • Microsoft • Oracle	BL	L4 – Analyse
Ex.	MySQL replication: primary handles writes, replicas handle read traffic. Asynchronous = better write performance.		

Q168	What is eventual consistency?		
Ans	In eventually consistent systems, replicas may return stale data temporarily but will converge to the same value given enough time with no new updates. Weaker than strong consistency but enables higher availability.		
Co.	Amazon • Google • Microsoft	BL	L4 – Analyse
Ex.	DNS propagation, DynamoDB, Cassandra — updates may take seconds to propagate to all nodes globally.		

Q169	What is the CAP theorem?		
Ans	A distributed database can only guarantee two of three simultaneously: Consistency (all nodes see same data), Availability (every request gets a response), Partition Tolerance (works despite network splits). Since partitions are inevitable, systems choose CP or AP.		
Co.	Amazon • Google • Microsoft • Oracle	BL	L5 – Evaluate
Ex.	HBase = CP (consistent+partition tolerant). Cassandra = AP (available+partition tolerant). Traditional RDBMS = CA (single node).		

Q170	Explain CAP Theorem.
Ans	CAP Theorem states distributed systems can only guarantee two of: Consistency, Availability, and Partition Tolerance simultaneously. Real-world systems must decide which two to prioritize.

Co.	Amazon • Google • Microsoft	BL	L5 – Evaluate
Ex.	DynamoDB: tunable consistency — can choose AP (default, higher availability) or CP (strong consistency, slower).		

Q171	What are database sharding and replication?		
Ans	Sharding = horizontal partitioning of data across multiple servers (each holds a subset). Replication = copying the same data to multiple servers (for availability, fault tolerance, and read scaling).		
Co.	Amazon • Google • Microsoft • Oracle	BL	L3 – Apply
Ex.	Sharding: writes distributed. Replication: reads distributed. Often used together for full horizontal scaling.		

12. Database Performance & Architecture

Q172	What is hot backup vs cold backup?		
Ans	Hot backup (online): taken while the DB is running and serving requests — uses WAL/logging. Cold backup (offline): taken when DB is shut down — simpler but requires downtime.		
Co.	Oracle • IBM • Amazon • Microsoft	BL	L2 – Understand
Ex.	Production DBs use hot backup (no downtime). Dev/test DBs use cold backup (simple copy while stopped).		

Q173	What is database partitioning? Horizontal vs. vertical?		
Ans	Partitioning splits a large table into smaller pieces. Horizontal: splits rows (e.g., by year). Vertical: splits columns (frequently used vs. rarely used columns in separate tables).		
Co.	Amazon • Google • Oracle • Microsoft	BL	L4 – Analyse
Ex.	Horizontal: Orders_2023, Orders_2024 partitioned by year. Vertical: Employee_Core(emp_id, name) + Employee_Extra(emp_id, bio, photo).		

Q174	What is connection pooling?		
Ans	A technique that maintains a cache of open database connections so new requests reuse existing connections instead of creating new ones. Creating a connection is expensive (TCP handshake, auth).		
Co.	Amazon • Google • Microsoft • Flipkart	BL	L4 – Analyse
Ex.	HikariCP (Java), pgBouncer (PostgreSQL) — without pooling, 1000 req/sec = 1000 new connections. With pooling: 20 reused.		

Q175	What is the N+1 query problem?		
Ans	Occurs when code fetches N records then issues one additional query per record — resulting in N+1 total queries. Fixed by using JOIN or eager loading.		
Co.	Amazon • Google • Flipkart • Zomato	BL	L5 – Evaluate
Ex.	Fetching 100 orders then querying customer name for each = 101 queries. Fix: JOIN orders with customers in 1 query.		

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Q176	What is database caching? How does it differ from indexing?		
Ans	Caching stores query results in fast memory (Redis, Memcached), avoiding the DB entirely. Indexing speeds up how the DB finds data on disk. Both are complementary — caching for repeated reads, indexing for efficient disk access.		
Co.	Amazon • Google • Flipkart • Zomato	BL	L4 – Analyse
Ex.	□ Redis caches homepage product list (microseconds). Index makes DB query for fresh data fast (milliseconds).		

Q177	What is relational algebra?		
Ans	A procedural query language for relational databases using set operators: SELECT (σ), PROJECT (π), JOIN (⋈), UNION (∪), INTERSECTION (∩), DIFFERENCE (−), RENAME (ρ).		
Co.	Oracle • IBM • TCS • Amazon	BL	L4 – Analyse
Ex.	□ π(name,salary)(σ(salary>50000)(Employee)) — project name and salary where salary > 50000.		

Q178	What is query processing in DBMS?		
Ans	Steps: Parsing (syntax check) → Translation (to relational algebra) → Optimization (cost-based best plan) → Execution Plan → Execution (actual retrieval). The query optimizer is key to performance.		
Co.	Amazon • Oracle • Google • Microsoft	BL	L4 – Analyse
Ex.	□ Query: SELECT * FROM Employees WHERE dept_id=3 AND salary>50000. Optimizer decides: use index on dept_id first vs. salary first.		

Q179	What is a covering index?		
Ans	An index that contains all columns required to satisfy a query — the DB engine never accesses the actual table rows, dramatically reducing I/O.		
Co.	Amazon • Google • Microsoft • Oracle	BL	L4 – Analyse
Ex.	□ Query: SELECT name, salary WHERE dept_id=3. Index on (dept_id, name, salary) = covering index — no table access needed.		

Q180	Explain locking in concurrency control.		
Ans	Shared (S) lock allows concurrent reads. Exclusive (X) lock for writes — blocks all other access. Two-Phase Locking (2PL): Growing phase (acquire all locks) + Shrinking phase (release all locks) — prevents many concurrency issues.		
Co.	Amazon • Oracle • Google • IBM	BL	L4 – Analyse
Ex.	□ Multiple users reading a report: S-lock (shared). Salary update: X-lock (exclusive — others wait until done).		

Q181	What is a hash join?		
Ans	A join algorithm using a hash table built from the smaller table, then probed with each row of the larger table. O(n+m) time — efficient for large tables without indexes on join columns.		
Co.	Amazon • Oracle • Google • Microsoft	BL	L5 – Evaluate

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Ex.	□ Hash join chosen by optimizer when no index exists on join column and both tables are large.

Q182	Difference between temporary table and table variable?		
Ans	Temporary table: session or procedure scope, supports indexes, statistics-aware, stored in tempdb. Table variable: batch scope, no non-clustered indexes, faster for small datasets.		
Co.	Microsoft • Oracle • Amazon	BL	L4 – Analyse
Ex.	□ Temp table for complex multi-step ETL processing. Table variable for simple lookups in a short stored procedure.		

Q183	What is MVCC (Multi-Version Concurrency Control)?		
Ans	DB keeps multiple versions of data items. Readers access a consistent snapshot from transaction start; writers create new versions. Reads and writes don't block each other.		
Co.	Amazon • Google • PostgreSQL • MySQL	BL	L5 – Evaluate
Ex.	□ PostgreSQL: long-running report query sees data snapshot from when it started, unaffected by concurrent inserts.		

Q184	What is WAL (Write-Ahead Logging)?		
Ans	All changes logged to disk before applied to data files. On crash, DB replays log to restore committed changes and undo uncommitted ones. Foundation of durability in ACID.		
Co.	Amazon • Oracle • PostgreSQL • Google	BL	L5 – Evaluate
Ex.	□ PostgreSQL WAL: INSERT writes to WAL first. Crash during write: WAL shows whether to commit or rollback on restart.		

Q185	What is a functional dependency? What are Armstrong's Axioms?		
Ans	FD X→Y means X uniquely determines Y. Armstrong's Axioms: Reflexivity (Y⊆X → X→Y), Augmentation (X→Y → XZ→YZ), Transitivity (X→Y, Y→Z → X→Z). These derive all valid FDs.		
Co.	Amazon • Oracle • Google • IBM	BL	L5 – Evaluate
Ex.	□ emp_id→name (reflexivity). emp_id→dept_id, dept_id→dept_name → emp_id→dept_name (transitivity).		